

Garbage Collectors

Every Hadamard test throws away half its data. We recycle it.

TBD-NAME TBD-NAME TBD-NAME

Qiskit Hackathon 2026 — Topic 5
after Faehrmann, Eisert & Kueng, *PRL* **135**, 150603 (2025)

14 August 2026

Outline

The garbage state

Does it work?

What each choice buys

Real hardware, and the price

Reproduce it, and where it breaks

Every number on these slides carries the RESULTS.md row that produced it, shown as [R0xx].

The garbage state

Does it work?

What each choice buys

Real hardware, and the price

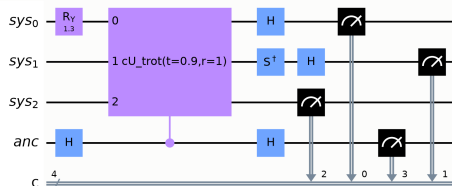
Reproduce it, and where it breaks

Every Hadamard test throws away half its data

The standard Hadamard test measures **one ancilla qubit** and **discards the system register**.

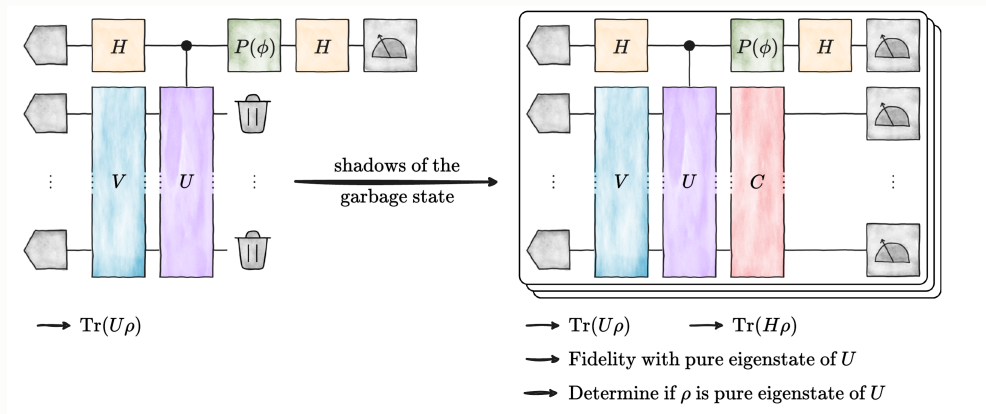
Measure that “garbage” in *random bases* instead and it becomes a **classical shadow** — the same shots now also carry the system’s observables.

We implement this end to end, validate every estimate against exact references, *quantify what it costs*, and characterise where it breaks.



3 system qubits + 1 ancilla

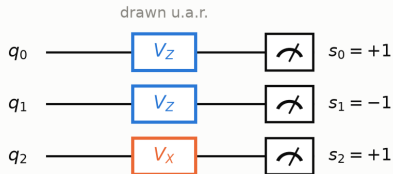
The protocol, in the authors' own picture



Left: the standard test — V prepares, c - U evolves, $P(\phi)$ picks the quadrature, and the system register goes **in the bin**. **Right:** append a random Clifford C and measure it instead: **same circuit, same shots**, now also $\text{Tr}(H\rho)$, fidelities and eigenstate tests.

What a classical shadow actually is

1. Each shot draws its own random basis



$b_j \sim \text{Unif}\{X, Y, Z\}$, independently on every qubit and every shot

All that is stored per shot: $(b_0 b_1 b_2, s_0 s_1 s_2)$

$$\hat{\rho} = \bigotimes_j \left(3 V_{b_j}^\dagger |s_j\rangle \langle s_j| V_{b_j} - \mathbb{1} \right), \quad \mathbb{E}[\hat{\rho}] = \rho$$

one unbiased snapshot — meaningless alone, exact in expectation once averaged

2. One record, every Pauli at once

$$\hat{\rho} = 3^{w(P)} \prod_{j \in \text{supp}(P)} s_j \mathbb{1}[b_j = P_j]$$

shot	basis b	outcome s	$\widehat{Z_0 Z_1}$	$\widehat{X_0 X_1}$
1	Z Z X	+ - +	-9	0
2	X X Z	+ + -	0	+9
3	Z Y Y	- + +	0	0
4	Z Z Z	+ + +	+9	0
5	Y X Z	- - +	0	0

$w = 2$: a shot lands with probability 3^{-2} and the 3^2 weight cancels it exactly, so $\mathbb{E}[\hat{\rho}] = \text{Tr}[\rho P]$

q_2 's basis never enters $\widehat{Z_0 Z_1}$: only $\text{supp}(P)$ must match
 $\Rightarrow 3^{-w}$, not 3^{-n} . The price is $\mathbb{E}[\hat{\rho}^2] = 3^w$.

No new circuits — only ≤ 2 extra single-qubit gates per qubit. The randomisation is the whole trick: it is what makes *one* record set an unbiased estimator for **every** Pauli observable simultaneously, instead of one chosen in advance. [Hsin-Yuan Huang, Kueng & Preskill, *Nat. Phys.* 16, 1050 (2020)]

One circuit, one set of shots, seven physical quantities

The ancilla alone gives the interference signal:

$$\langle Z_a \rangle_\phi = \operatorname{Re} \left[e^{i\phi} \operatorname{Tr}(U\rho) \right], \quad \phi = 0 \rightarrow \operatorname{Re} \chi, \quad \phi = -\frac{\pi}{2} \rightarrow \operatorname{Im} \chi$$

Tracing out the ancilla leaves the *unweighted* shadow target, and weighting each snapshot by the ancilla outcome $a = \pm 1$ leaves the *joint* one:

$$\rho^{(I)} = \frac{1}{2}(\rho + U\rho U^\dagger), \quad \mathbb{E}[a \hat{P}] = \operatorname{Re} \left[e^{i\phi} \operatorname{Tr}(P U\rho) \right]$$

From **one** record set: $\chi(t)$, $\chi_Q(t)$, $\chi_H(t)$, $\chi_{Z_0}(t)$, $\langle H \rangle$, $\langle Q \rangle$, $\langle Z_0 \rangle$ [R009]

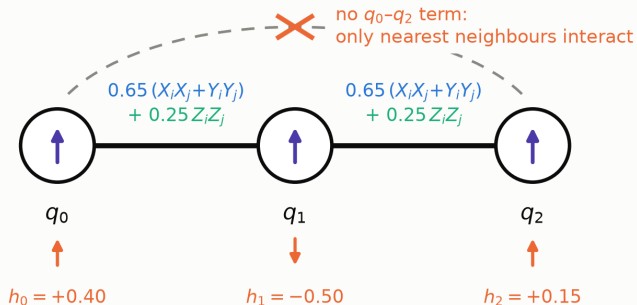
3,456 circuits · 256,000 shots — one experiment, not seven.

The benchmark: three spins, nearest-neighbour coupling only

3 spin- $\frac{1}{2}$ particles, open chain

conserved: $Q = \sum Z_j$, $[H, Q] = 0$

$\Rightarrow$ the spectrum splits into charge sectors



the three local fields are unequal, so no two levels coincide

$$H = \sum_{i=0}^1 [0.65 (X_i X_{i+1} + Y_i Y_{i+1}) + 0.25 Z_i Z_{i+1}] + 0.40 Z_0 - 0.50 Z_1 + 0.15 Z_2$$

Why this model. Small enough to diagonalise exactly, so every estimate has a truth reference — yet it still carries both features the protocol exploits: a **conserved charge** and a **non-degenerate spectrum**.

The initial state $R_y^{(q_0)}(1.3) |000\rangle$ reaches **4 of the 8 levels**:

$E \in \{-1.9152, -0.4810, +0.5500, +1.9463\}$ in sectors $q = +1, +1, +3, +1$. [R002, R012]

The garbage state

Does it work?

What each choice buys

Real hardware, and the price

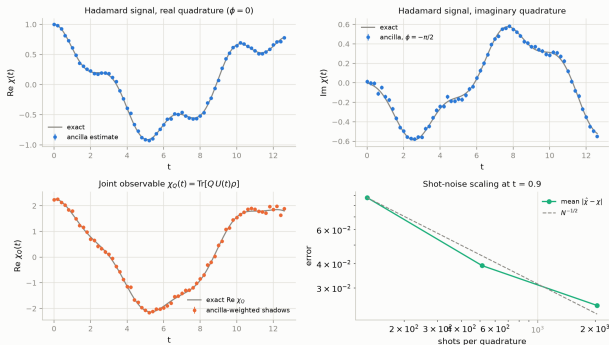
Reproduce it, and where it breaks

Every estimator agrees with theory — and the error bars are honest

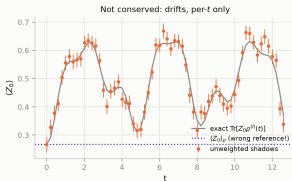
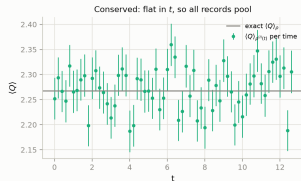
- **56/56 checkpoints pass**, zero hard failures, zero advisory warnings.
- $\chi(t)$ rms error **0.0290** vs predicted sem **0.0277** — ratio **1.05**, so the error bars neither over- nor under-state. [R010]
- Shot-noise slope **-0.463** vs theory -0.5 : unbiased, statistics-limited. [R010]

The single-shot second moment is exactly $3^{w(P)}$, which predicts the measured error bars to **within 2%**. [R029]

Fundamental goal: shadow-enhanced Hadamard test vs. exact reference



Conserved quantities come out flat — from the very same shots



$$\langle H \rangle = +0.0714 \pm 0.0082$$

exact +0.0739

$$\langle Q \rangle = +2.2708 \pm 0.0052$$

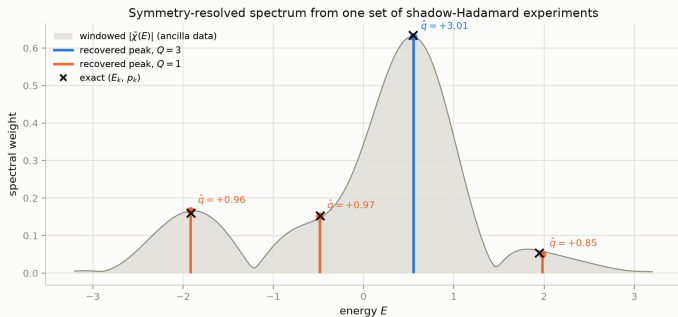
exact +2.2675

Pooled over all 128 settings, extracted from the *discarded* register of the $\chi(t)$ experiment. [R010]

Per-time $\langle Q \rangle$ is flat within 5σ at every one of the 64 time points (max 2.2σ). [R010]

No extra circuits (≤ 2 extra 1-qubit gates per system qubit); the statistical cost is on slide 20.

We recover the spectrum *and* label each line by symmetry sector



All 4 populated levels recovered:
 $\max |\Delta E| = \mathbf{0.033}$, $\max |\Delta p| = \mathbf{0.006}$, **every label correct**. [R012, R013]

Reproducible: **12/12 independent seeds** got all four labels right. [R019]

The colour is the deliverable. It is information the grey curve does not contain — and it cost *no additional circuits*.

The grey band is the plain windowed Fourier transform. A better estimator sharpens it; *no* estimator can colour it.

The garbage state

Does it work?

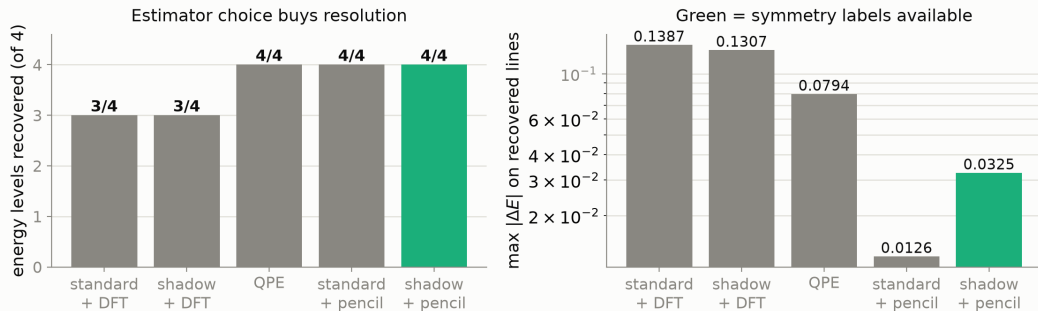
What each choice buys

Real hardware, and the price

Reproduce it, and where it breaks

What each choice actually buys

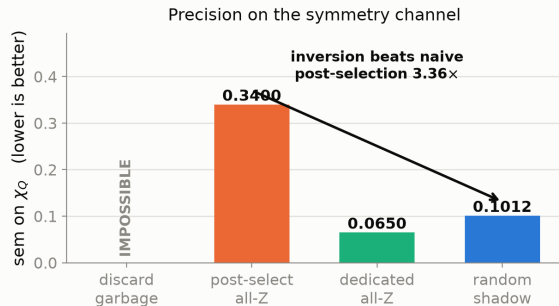
Only the shadow protocol yields symmetry labels — at any shot count



Estimator choice buys resolution (3/4 → 4/4 lines). **Protocol** choice buys the labels — and nothing else can. [R011, R012, R021, R030]

Do you even need shadows? Four ways to treat the garbage

Four ways to treat the garbage register — each rung buys something



What each can measure AT ALL

protocol	$\chi(t)$	Q	H terms
discard garbage	yes	--	--
post-select all-Z	yes	yes	5/9
dedicated all-Z	yes	yes	5/9
random shadow	yes	yes	9/9

Z-only protocols are blind to the 4 hopping terms (0.65 each):
(H) is conserved, yet what they report drifts by 0.147
— with no second channel to notice.

Each rung buys a capability the one below *cannot have at any shot count*. [R039, R040]

What the shadow inversion actually contributes

$Q = \sum_j Z_j$ is diagonal in the computational basis, so a **dedicated Z readout beats our shadow estimator by $1.56\times$** on the symmetry channel. We report that. [R039]

But compare the shadow inversion against *post-selecting the all-Z shots out of its own data*:

$$\underbrace{\frac{1}{3}}_{\text{per term } Z_j: b_j=Z} \quad \text{versus} \quad \underbrace{\frac{1}{3^n}}_{\text{all qubits } Z \text{ at once}} = 3.7\% \text{ of shots}$$

The inversion beats naive post-selection of the same data by $3.36\times$ (sem $0.3400 \rightarrow 0.1012$) — it recovers exactly the shots post-selection discards. [R040]

The case for shadows is not better symmetry resolution.

It is that the *same* shots also give you H — which no Z-only protocol can see.

Which protocol can rebuild which observable?

observable	what it measures physically	discard	post-select all-Z	shadow
		Hadamard $\rightarrow$ DFT	$\rightarrow$ matrix pencil	$\rightarrow$ matrix pencil
energies	the level structure of H	3/4 lines	4/4	4/4
Z_i	local magnetisation of spin i	—	yes (err 0.41 at 3.7% yield)	yes (0.048)
$Z_i Z_j$	neighbours aligned vs anti-aligned	—	yes (err 0.28)	yes (0.057)
H	total energy	—	biased (5/9 terms, 2.70)	yes (0.139)
$X_i X_j + Y_i Y_j$	hopping : excitation exchanged between i, j	—	impossible (0/2 terms)	yes (0.509)

A Z-only protocol sees only what is *diagonal*: it can say how the spins are *oriented*, never how they *move*. The hopping term is precisely the part of H that transports an excitation along the chain — the part that makes the dynamics non-trivial — and it converges to $\text{Tr}[O_{Z\text{-diag}} U \rho]$, so more shots do not help.

The shadow's exclusive capability is non-commuting observables. [R041]

The garbage state

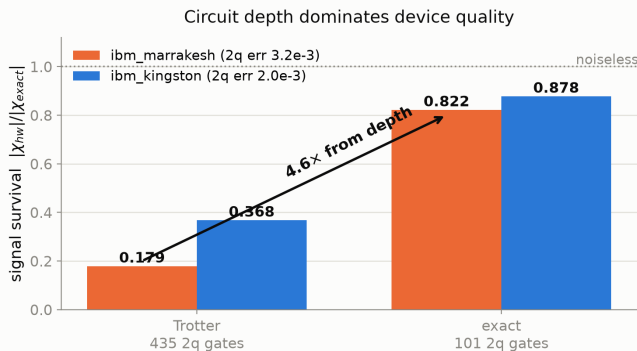
Does it work?

What each choice buys

Real hardware, and the price

Reproduce it, and where it breaks

It survives on real hardware once the circuit is shallow enough



Complete 2×2 grid. Depth dominates ($2.4\text{--}4.6\times$) over device choice ($1.1\text{--}2.1\times$). [R008, R018]

With a *full random-basis ensemble* on hardware: χ survival **0.962**, and a valid $\langle Q \rangle$ at **0.957** — the core claim, on a real QPU. [R023]

Caveat, stated plainly: one run per configuration. An independent run of an identical circuit 30 minutes later differed by $> 2\times$. Hardware varies; we report that. [R026, R031]

The honest cost: 13 experiments replaced for a $1.07\times$ variance premium

The per-shot Pauli estimator and its exact second moment:

$$\hat{P} = 3^{w(P)} \prod_{j \in \text{supp}(P)} s_j \mathbf{1}[b_j = P_j], \quad \mathbb{E}[\hat{P}^2] = 3^{w(P)}$$

Conventional route: **13 dedicated** modified Hadamard tests ($Q=3$, $H=9$, $Z_0=1$ Pauli terms).

Shadow route: **0 extra circuits** (≤ 2 extra 1q gates per qubit).

Price paid instead — measured, not asserted:

$$\frac{\text{sem}_{\text{shadow}}}{\text{sem}_{\text{dedicated}}} = \frac{0.1012}{0.0949} = \mathbf{1.07\times}$$

on χ_Q specifically. [R021]

The 3^w model predicts the observed error bars to **within 2%**. [R029]

The garbage state

Does it work?

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Reproduce it, and where it breaks

Reproduce it in under five minutes

- **37 sourced results rows** — every number maps to the command that produced it (EVIDENCE.md).
- Quickstart reproduces our headline *table* from saved data; the money plot ships pre-rendered.
- `hardware_run.py` reproduces our hardware arm in 3 commands.
- `BUGLOG.md` — including the numbers we got wrong and retracted.

First open implementation of PRL **135**, 150603. Prior-art search 2026-08-13; scope and limits recorded in `deck/PRIOR_ART_SEARCH.md`.

QR CODE

TBD-NAME: paste repo QR here

What we could not do, and where it breaks

- The windowed DFT **never** resolves the 1.031-spaced pair at *any* $T_{\max}$ we tested; the matrix pencil resolves it from $T_{\max} \geq 7.0$. A method limit, not a shot-count limit. [R011, R022]
- Exact-synthesis shallowness is an **$n=3$ artefact** — the advantage falls $3.41\times \rightarrow 1.16\times$ from $n=3$ to $n=4$ and is *expected* to invert beyond (measured at $n=3, 4$ only). Not a scalable recipe. [R025]
- One noise study came out **confounded**; we exclude it rather than report it. [R017]
- Hardware supports *robustness claims only* — never a headline number. All Part A/B results are simulator.

We make no claim of outperforming classical computation.

The system is 3 qubits; a laptop diagonalises it instantly. It is small *on purpose*, so every estimate can be graded against an exact answer.

Team

- **TBD-NAME** — TBD-CONTRIBUTION
- **TBD-NAME** — TBD-CONTRIBUTION
- **TBD-NAME** — TBD-CONTRIBUTION

Garbage Collectors

Thank you — questions welcome.

Backup slides

Backup — the standard and shadow protocols give *identical* χ

Measured, not assumed: a genuine textbook Hadamard test over the full 64-point grid at identical budget gives

$$\text{rms}_{\text{std}} = 0.0265, \quad \text{rms}_{\text{shadow}} = 0.0290, \quad \text{mean sem } 0.0277 \text{ vs } 0.0277$$

with $|\chi_{\text{std}} - \chi_{\text{shadow}}|/\text{combined sem}$: max 2.10σ , mean 0.88σ — statistically the **same random variable**. [R021]

So measuring the garbage register costs the ancilla signal *nothing*, and the two axes factorise cleanly: estimator $\rightarrow$ resolution, protocol $\rightarrow$ labels.

Backup — a self-validating decoherence witness

For a pure input state, exactly:

$$\mathrm{Tr}[(\rho^{(I)})^2] = \frac{1}{2} (1 + |\chi(t)|^2)$$

Left side from the **system register alone** (shadow snapshot pairs); right side from the **ancilla alone**. Two independent channels of the same experiment, so their gap certifies decoherence with **no exact diagonalisation and no classical simulation** — it survives to sizes where nothing can be verified.

ideal simulator	-0.1σ	(correct null)
noisy sim, $1\times$		-7.6σ (fires)
noisy sim, $5\times$	-17.2σ	(dose-response)
real hardware (cleanest run)	-1.4σ	(correct null)

[R033]

Backup — future work, already prototyped: density of states

The DOS is the Fourier transform of the *trace* of the propagator:

$$G(t) = \frac{1}{\sqrt{2\pi}} \text{Tr} \left[e^{-i\hat{H}t} \right] \quad (\text{Goh \& Koczor, arXiv:2407.03414v2, Eq. 7})$$

Our $\chi(t) = \text{Tr}[U(t)\rho]$. Set $\rho = \mathbb{I}/2^n$ and **the same experiment becomes a DOS estimator** — one line changed, analysis chain untouched.

Prototyped: peak weights correctly become *degeneracies* (isolated levels 0.11–0.13 vs predicted 1/8; a 3-fold merged peak 0.354 vs predicted 3/8). [R038]

Where we could improve on that paper: it obtains fixed-particle-number DOS with *one experiment per sector*. Our shadow channel measures $\langle Q^k \rangle$ from the *same* shots, so the charge moments at each peak recover **all sectors from a single run**. Feasibility checked: Q^2, Q^3 are 4 Pauli terms each; moment inversion condition number 31.

Backup — two bugs we caught, and how

B04 — a fixed-basis shadow estimator silently returns a *different* observable. The estimator is unbiased only for i.i.d.-uniform bases; with one fixed basis the indicator is deterministic and non-matching terms contribute structurally zero. We reported a hardware $\langle Q \rangle$ this way and blamed noise. On a *noiseless* simulator the same code is **65 σ off** — so it was never noise. Retracted; `hardware_run.py` now refuses to print system observables without `--shadow`.

B06 — an error bar computed over the wrong units. Purity is a U-statistic over *pairs* of snapshots; we used $\text{std}/\sqrt{n_{\text{pairs}}}$, but pairs share shots, so the effective sample size is the number of *shots*. That faked a 2.2 σ signal. **Only the ideal-simulator control leg exposed it** — the hardware number alone looked fine.

A witness validated only where the truth is unknown cannot be trusted.